

WASP-2b

R-2408

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_110901 · created 2026-08-21T01:12:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2455805.8423 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.047
Transit depth (Rp/Rs)^2	2.5 +/- 1.49 [%]
Orbital Inclination (inc)	89.46 +/- 2.87
Airmass coefficient 1 (a1)	1.79 +/- 0.78
Airmass coefficient 2 (a2)	-0.35 +/- 0.19
Scatter in the residuals of the lightcurve fit is	42.03 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.75
Transit Duration (day)	0.092 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.047 vs 0.1326 (deviation 0.54σ)
Duration fit vs published	0.092 d vs 0.0737 d (deviation 1.4σ)
Mid-time O–C	5.4 min
Depth SNR	3.4
Residual scatter	42.03 %

Light curve

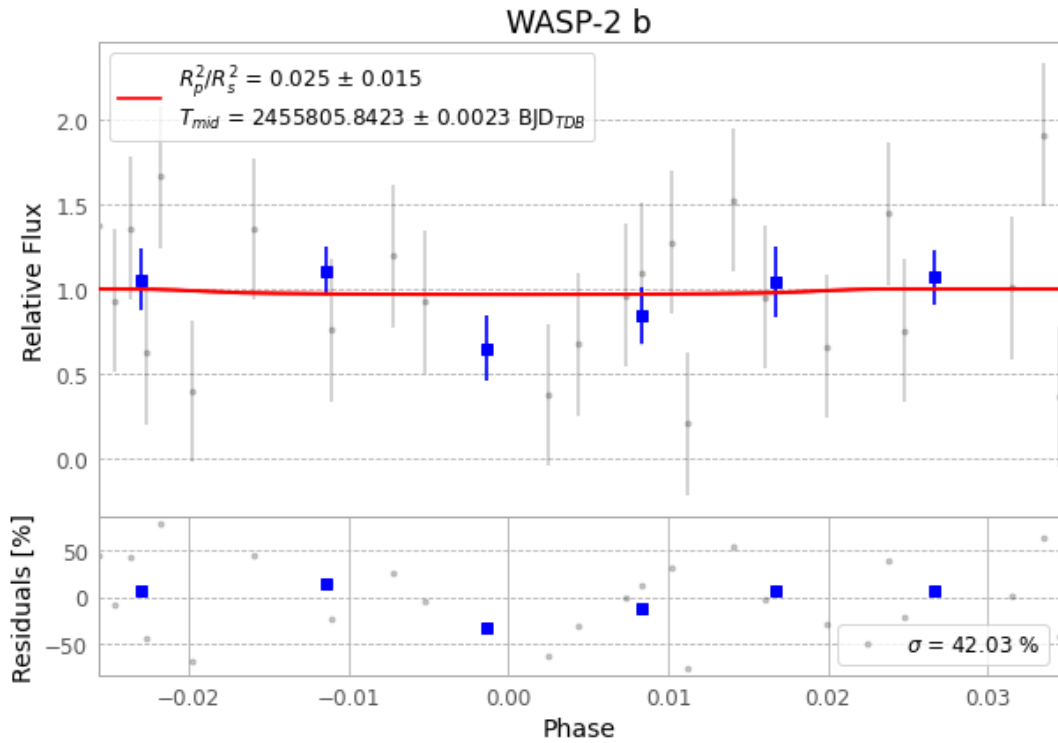


Figure 1 — FinalLightCurve_WASP-2 b_2011-08-31.png (SHA-256 850228065e4d8adb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [446, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a01285740d602b96...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), WASP-2110901064213.FITS → WASP-2110901100612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-110901.log (c7f3e8e885eb...), FinalLightCurve_WASP-2 b_2011-08-31.png (850228065e4d...), FinalLightCurve_WASP-2 b_2011-08-31.csv (c5689cdd633d...)

Compute resources

Wall time 143.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 01:12Z.